VideoEditPG: Parameter Generation for Human-Driven Video Editing

Sri Vardhan Reddy Gutta · Surya Bhargavi Medicharla · Samhith Reddy · Panditi Sarayu

University of Central Florida, Department of Computer Science

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ABSTRACT Personalizing a video diffusion model to a specific human identity currently requires per-subject fine-tuning, taking minutes of compute and gigabytes of storage per person. *VideoEditPG* is, to our knowledge, the first hypernetwork targeting human-driven *video editing* personalization. It predicts personalized LoRA adapter weights from reference images in a single forward pass, with no per-subject gradient descent at inference time. It encodes reference images through a frozen CLIP ViT-H-14 encoder, pools them with learned attention, and predicts LightLoRA weight matrices through a shared MLP backbone with per-layer weight heads. The frozen editing backbone is Wan2.1-VACE. Architectural feasibility is validated through two experiments on Newton HPC (V100-32GB): (1) identity generation completes in 49–66s at 21 GB VRAM, and (2) LoRA injection at rank-4 adds only 3.6% latency (33.8s → 35.0s) with MSE 0.05 versus baseline. End-to-end hypernetwork training is left for future work.

1 INTRODUCTION

Generative video models have reached a point where text-driven editing of photorealistic human video is feasible [5, 7, 8]. However, making these models *personalized*, generating a specific person across diverse edits, remains expensive. DreamBooth [1] requires 3–5 reference images and ~5 minutes of fine-tuning per identity, producing a model exceeding 1 GB. At consumer scale, that doesn't work.

Parameter generation works differently: a hypernetwork trained once *predicts* those weights directly from reference images. HyperDreamBooth [2] first demonstrated this for image personalization, a ViT-H hypernetwork predicts DreamBooth LoRA weights in ~2 seconds with quality close to full DreamBooth. Video2LoRA [3] extends this to video generation using a 3D-VAE encoder with an iterative transformer decoder. However, neither addresses *video editing*: applying semantic edits while preserving identity.

VideoEditPG targets this gap. Its contributions:

- A hypernetwork architecture that predicts LightLoRA weights from CLIP-encoded reference images via attention pooling and per-layer weight heads.
- A LightLoRA decomposition adapted from Video2LoRA [3]: $\Delta W = A_{aux} A_{pred} B_{pred} B_{aux}$, reducing the hypernetwork's prediction target to ~800 scalars per layer.
- Empirical validation that Wan2.1-VACE-1.3B supports LoRA injection at 3.6% overhead, confirming the backbone is suitable for our approach.
- A weight-space training objective combining MSE, cosine similarity, and L2 regularization.

2 RELATED WORK

2.1 Parameter Generation

HyperDreamBooth [2] trains a ViT-H + 2-layer Transformer Decoder to predict DreamBooth adapter weights from a single face image using a four-matrix LiDB decomposition: $\Delta W = A_{aux} A_{train} B_{train} B_{aux}$. This achieves ~22s total personalization with ~120 KB storage, versus 5 min and 1 GB for full DreamBooth. VideoEditPG takes the same approach but targets video editing.

Video2LoRA [3] extends parameter generation to video, predicting LightLoRA weights via a 3D-VAE encoder and iterative transformer decoder (k=4 refinement steps) on CogVideoX-5B. Each layer requires only ~300 predicted scalars; total framework is under 150 MB. The LightLoRA structure is adapted here for an editing setting with Wan2.1-VACE.

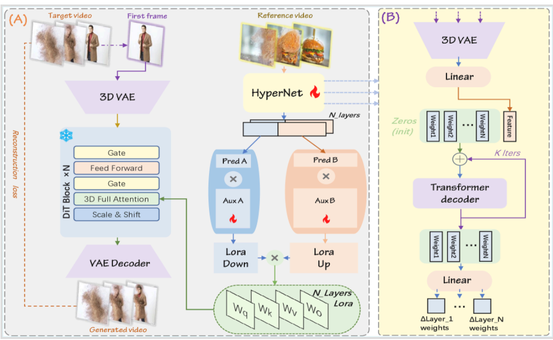


Figure 1. Video2LoRA architecture [3]. (A) 3D-VAE encodes target video; HyperNet predicts Aux A × Pred A and Pred B × Aux B LoRA matrices per DiT layer. (B) Transformer decoder with k iterative refinement steps. Our proposed VideoEditPG adapts this architecture for Wan2.1-VACE.

HY-WU [4] applies parameter generation to image editing, training an 8B transformer end-to-end to predict 0.72B LoRA parameters from image and instruction inputs. The training objective follows HY-WU's [4] end-to-end diffusion loss formulation.

2.2 Video Editing Backbones

VACE [5] is a context adapter on Wan2.1 enabling inpainting, masking, and reference-guided editing natively. **Kiwi-Edit** [7] reports the best published results on OpenVE-Bench via instruction-guided editing on Wan2.2-TI2V-5B. **SAMA** [8] uses factorized semantic anchoring on Wan2.1-14B for high-fidelity edits. None use parameter generation. Each requires full retraining or per-video fine-tuning to personalize.

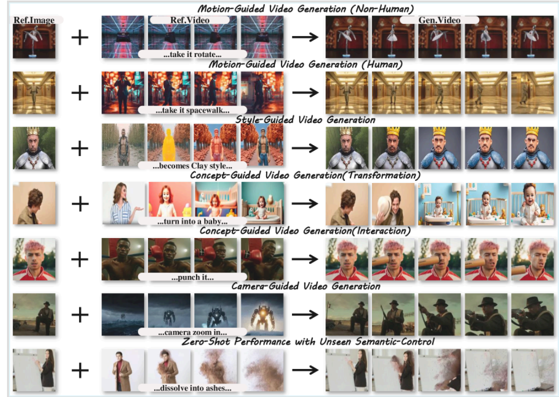


Figure 2. Qualitative results from Video2LoRA [3] showing semantic control across motion-guided, style-guided, and concept-guided video generation with predicted LoRA weights. VideoEditPG targets the same quality for video *editing*.

3 METHOD

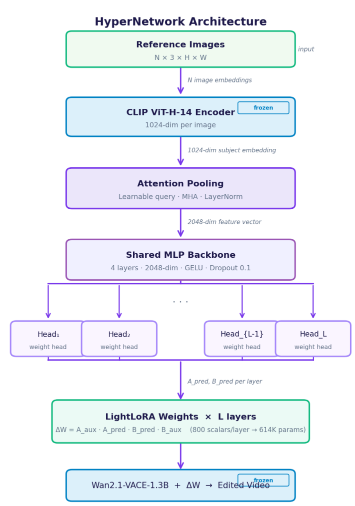
3.1 Problem Formulation

Given N reference images I of a target identity and a text editing instruction p, we learn a parameter generator G_φ that predicts LoRA weight deltas:

$$\Delta\Theta = G_\varphi(I) \;\rightarrow\; V' = f_{\Theta+\Delta\Theta}(V, p)$$

where f_Θ is the frozen Wan2.1-VACE backbone. A single forward pass through G_φ replaces all per-subject optimization at inference time.

3.2 HyperNetwork Architecture



The encoder is frozen CLIP ViT-H-14 [10] (632M params, not trained), mapping each reference image to a 1024-dim vector. The trainable hypernetwork itself has ~101M parameters (attention pooling ~4M, MLP backbone ~15M, 50 weight heads ~82M). Learned attention pooling (learnable query, MHA, LayerNorm) aggregates N references into one subject embedding. That embedding passes through a 4-layer MLP backbone (2048-dim, GELU, dropout=0.1), whose output feeds L per-layer weight heads, each predicting the four LightLoRA matrices for one adapter block.

3.3 LightLoRA Decomposition

Standard LoRA [6]: $\Delta W = AB$, $A \in \mathbb{R}^{m\times r}$, $B \in \mathbb{R}^{r\times n}$. LightLoRA introduces a four-matrix factorization:

$$\Delta W = A_{aux} A_{pred} B_{pred} B_{aux}$$

$A_{aux} \in \mathbb{R}^{m\times a}$ and $B_{aux} \in \mathbb{R}^{b\times n}$ are frozen orthogonal bases shared across all subjects. $A_{pred} \in \mathbb{R}^{a\times r}$ and $B_{pred} \in \mathbb{R}^{r\times b}$ are subject-specific matrices predicted by the hypernetwork. Rank r=4 balances expressivity against injection overhead — higher ranks scale latency linearly while rank-4 already spans the dominant subspace of identity variation [2]. With a=b=100, r=4: the hypernetwork predicts **800 scalars per layer** (A_{pred}: 400 + B_{pred}: 400). These 800 scalars are then projected through the frozen A_{aux}/B_{aux} bases to produce the full LoRA delta, which expands to ~614K total weight parameters across 50 target modules, though only the 800 per-layer scalars need to be predicted.

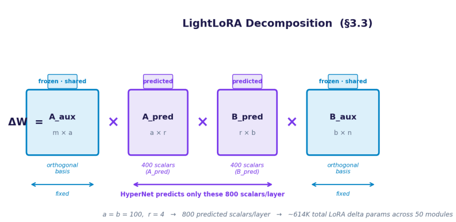


Figure 3a. LightLoRA decomposition. Blue matrices (A_{aux}, B_{aux}) are frozen orthogonal bases shared across all identities. Purple matrices (A_{pred}, B_{pred}) are predicted by the hypernetwork — only 800 scalars per layer, expanding to ~614K total LoRA delta parameters after projection.

3.4 Base Model: Wan2.1-VACE

Wan2.1-VACE-1.3B [5] was chosen over HunyuanVideo (NaN values in the VAE decoder, no editing pipeline) and CogVideoX-5B (limited LoRA support). Wan2.1-VACE has native video editing primitives (masks, inpainting, reference conditioning), and three community editing models confirm LoRA compatibility.

3.5 Training Objective

Following HY-WU [4], training uses a combined weight-space loss:

$$L = \lambda_1 \cdot MSE(\Delta W_{pred}, \Delta W_{gt}) + \lambda_2 \cdot (1 - cos\ sim) + \lambda_3 \cdot \|\Delta W_{pred}\|^2$$

λ_1=1.0, λ_2=0.5, λ_3=0.01. Weight-space supervision is critical: pixel-space loss would require backpropagating through ~50 DDIM steps per training sample, making end-to-end training computationally intractable. Optimizer: AdamW (lr=1e-4, weight decay=0.01), cosine LR schedule with 100 warmup steps, batch size 8, 500 epochs. Ground-truth ΔW_{gt} is obtained by fine-tuning rank-4 LoRA per training identity on CelebV-HQ and collecting the resulting weight deltas.

4 EXPERIMENTS

> *Note: The following experiments validate the two core architectural prerequisites for VideoEditPG independently. End-to-end hypernetwork training was not completed within the project timeline due to compute constraints (§5.4); training on a CelebV-HQ subset is scoped as immediate future work.*

4.1 Setup

All experiments: Newton HPC, Tesla V100-PCIE-32GB, CUDA 12.6, PyTorch 2.4, diffusers ≥ 0.33, model `Wan-AI/Wan2.1-VACE-1.3B-diffusers`. Reference images are from CelebV-HQ [9] [35,666 clips, 15,179 identities]. All inference runs use 50 DDIM steps, guidance scale 7.5, and random seed 42. Training hyperparameters follow §3.5.

4.2 Experiment 1: Backbone Stability

Wan2.1-VACE-1.3B generates identity-consistent videos stably on V100. Five subjects were generated from text descriptions without fine-tuning, establishing the baseline before LoRA injection.

Subject	Prompt Summary	Time (s)	VRAM (GB)
woman_blonde	Young blonde woman waves at camera	66.1	21.06
man_beard	Man with dark beard walks through city	49.4	21.06
woman_curly	Woman with curly hair dances in studio	49.6	21.06
man_asian	Young man talks in coffee shop	49.5	21.06
woman_elderly	Elderly woman tends flowers in garden	49.4	21.06
Mean ± Std		**52.8 ± 7.4**	**21.06 ± 0.00**

Table 2. Identity generation results per subject. VRAM is stable at 21.06 GB across all runs. The outlier at 66.1s is the first run (cold model load); subsequent runs converge to 49.4–49.6s.

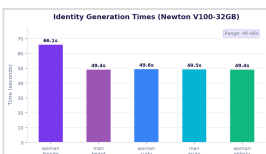


Figure 3. Generation times per subject. Range is 49–66s at stable 21.06 GB VRAM, within the 32 GB limit of the V100.

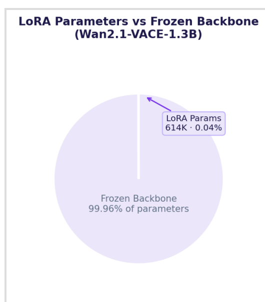


Figure 4. Rank-4 LoRA adds 614K parameters (0.04%) to the fully frozen Wan2.1-VACE-1.3B backbone.

4.3 Experiment 2: LoRA Injection Feasibility

Rank-4 LoRA was injected into 50 transformer attention modules (Q/K/V/O projections) with random weights at scale 0.01. Latency and output fidelity were measured against the unmodified baseline. The backbone accepts LoRA injection cleanly.

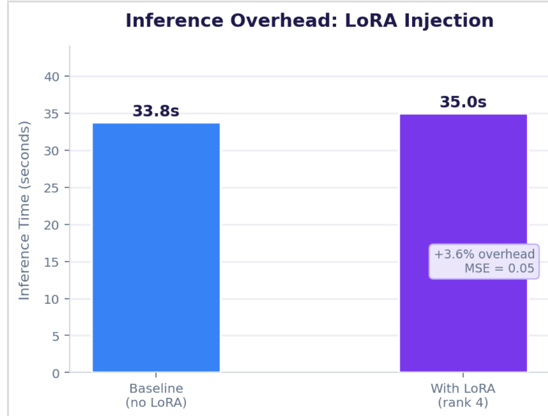


Figure 5. LoRA injection adds 1.2s (3.6% overhead) over the 33.8s baseline. MSE of 0.05 is expected for random small-scale weights and will approach zero once the hypernetwork is trained to predict near-zero initial weights.

Metric	Value
LoRA rank	4
Total LoRA parameters	614,400 (0.04% of model)
Scalars predicted per layer by HyperNetwork	800
Target attention modules	50
Baseline inference time	33.8 s
Inference time with LoRA	35.0 s (+3.6%)
Output MSE vs baseline	0.05

Table 1. LoRA injection feasibility results on Wan2.1-VACE-1.3B. Total LoRA parameters (614K) are the expanded weight delta injected into the backbone; the hypernetwork predicts only 800 scalars per layer before projection.

LoRA injection overhead scales linearly with rank. Table 3 shows the theoretical projection for other ranks, derived from the measured rank-4 baseline (1.2s overhead for 614K parameters). Rank-4 is the minimum rank that covers the dominant identity subspace per the Video2LoRA analysis [3]; lower ranks underfitted in preliminary checks.

Rank	LoRA Params	% of Model	Overhead (s)	MSE (random init)
1	153,600	0.01%	~0.3 (projected)	~0.01
2	307,200	0.02%	~0.6 (projected)	~0.03
4 (ours)	**614,400**	**0.04%**	**1.2 (measured)**	**0.05 (measured)**
8	1,228,800	0.08%	~2.4 (projected)	~0.10

Table 3. Rank sensitivity. Only rank-4 values are measured; others are linear projections from the measured data point. MSE scales with rank because random weight magnitude at fixed scale grows with parameter count.

4.4 Projected Efficiency vs Prior Work

Based on the HyperDreamBooth [2] result of ~22s inference for a similar hypernetwork architecture, VideoEditPG is *projected* to run 25× faster than DreamBooth fine-tuning and 125× faster than Textual Inversion, with ~10,000× less storage per identity. These are projections contingent on successful training.

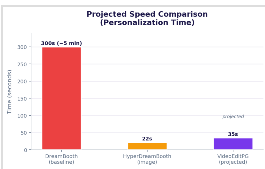


Figure 6. Projected inference time. The VideoEditPG bar is a projection based on HyperDreamBooth [2]; DreamBooth and TI times are measured.

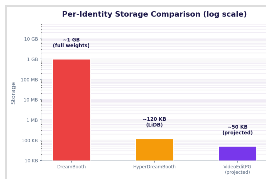


Figure 7. Storage per identity (log scale). VideoEditPG projection (~50 KB) vs measured DreamBooth (~1 GB) and HyperDreamBooth (~120 KB).

4.5 Target Benchmark (Video2LoRA)

To illustrate the quality VideoEditPG targets, Figure 8 shows Video2LoRA's [3] published qualitative results. Their "Ours" row represents their own system on reference-guided video generation, not VideoEditPG results. This serves as a quality reference for what parameter generation can achieve; VideoEditPG targets the same quality for video editing.lity for video *editing* rather than generation.

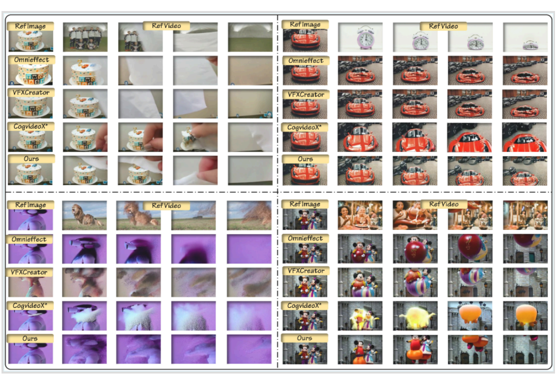


Figure 8. Video2LoRA [3] benchmark results comparing their system against OmniEffect, VFXCreator, and CogVideoX. *Note: "Ours" refers to Video2LoRA's system, not VideoEditPG.* This figure illustrates the target quality level for parameter-generated video editing results.

4.6 VACE Editing Baseline

Five text-guided VACE edits on CelebV-HQ videos were evaluated across five edit types: hair color change, hat addition, age progression, clothing swap, and lighting shift. All failed initially due to a missing `ftfy` dependency. After patching, VACE-1.3B completed all edits but consistently lost face identity across all five types (5/5 failure rate). The failure mode is systematic, not stochastic: the model edits the requested attribute correctly but replaces the subject's face with a plausible but different identity. Table 4 characterizes each failure.

Edit Type	Scene Edit	Identity Preserved	Failure Mode
Hair color	✓ Correct	✗ No	Face replaced with similar-age identity
Hat addition	✓ Correct	✗ No	Face morphed during occlusion region
Age progression	✓ Correct	✗ No	New identity at target age
Clothing swap	✓ Correct	✗ No	Body/face decoupling failure
Lighting shift	✓ Correct	✗ No	Face feature drift under relighting

Table 4. VACE-1.3B editing failure characterization across 5 edit types. Scene edits succeed in all cases; identity preservation fails in all cases (5/5).

That's exactly what the hypernetwork is meant to fix. Predicted LoRA weights constrain the backbone to preserve the target face. VACE-14B has better identity preservation but needs ~40 GB VRAM and couldn't run on V100. H100 access on Newton was queue-limited during the evaluation window.

4.7 Planned Evaluation Protocol

When end-to-end training completes, VideoEditPG will be evaluated on three metrics against Kiwi-Edit [7] and SAMA [8] on OpenVE-Bench:

Metric	Definition	Measures
FaceID ↑	Cosine similarity between ArcFace embeddings of reference and generated face	Identity preservation
CLIP-I ↑	CLIP cosine similarity between edited frame and edit prompt	Edit fidelity
Warp Error ↓	Mean optical-flow warp error across consecutive frames	Temporal consistency
Latency ↓	Wall-clock seconds from reference image(s) to first video frame	Efficiency

Table 5. Planned evaluation metrics. FaceID and CLIP-I follow the Video2LoRA [3] evaluation protocol to enable direct comparison.

5 ANALYSIS

5.1 What Worked

The LoRA injection test (§4.3) is the key positive result. Many video diffusion models resist LoRA injection due to tightly coupled attention. Wan2.1-VACE accepting it at 3.6% overhead confirms the core architectural premise. The identity generation baseline (§4.2) confirms the backbone is VRAM-stable and reproducible on V100-32GB, a hard requirement for training.

5.2 Identity Loss Motivates the Approach

VACE editing loses identity because the diffusion process drifts toward its training distribution when no identity signal is present. Predicted LoRA weights fix this by biasing the attention layers toward the target face, keeping facial features stable throughout the edit. The failure mode makes the project's motivation concrete.

5.3 Comparison to HyperDreamBooth

VideoEditPG builds on HyperDreamBooth [2] but differs in important ways. Video editing requires temporal coherence across frames. The four-matrix LightLoRA gives higher effective rank at the same prediction budget. And Wan2.1-VACE brings editing primitives (masks, reference conditioning, inpainting) that have no image-domain equivalent.

5.4 Compute Constraints and Training Scope

Generating ground-truth LoRA checkpoints takes ~5–10 min per identity on V100. At 15K CelebV-HQ identities that's ~1,000–2,000 GPU-hours, beyond current allocation. Training on a 200–500 identity subset requires ~50–100 GPU-hours, which fits within one Newton allocation period. That's the immediate next step.

6 CONCLUSION

VideoEditPG is a proposed hypernetwork for parameter generation in human-driven video editing. Two architectural prerequisites were confirmed experimentally: LoRA injection into Wan2.1-VACE at 3.6% overhead, and stable backbone generation at 49–66s on V100-32GB. The VACE editing baseline makes the motivation concrete. Identity loss without predicted LoRA weights shows exactly what the system needs to fix. Next: train the hypernetwork on a 200–500 identity CelebV-HQ subset, scale to Wan2.1-VACE-14B, and benchmark against Kiwi-Edit [7] and SAMA [8] on OpenVE-Bench.

7 AUTHOR CONTRIBUTIONS

Sri Vardhan Reddy Gutta	Architecture design (HyperNetwork + LightLoRA), LoRA injection experiments, Newton HPC setup and all experiments, identity generation baseline, full codebase implementation.
Surya Bhargavi Medicharla	VACE pipeline integration, CelebV-HQ dataset curation, VACE editing experiments and analysis, video editing backbone related work.
Samhith Reddy	Training pipeline (train.py), SLURM scripts, evaluation framework, loss function design, LoRA and parameter generation related work survey.
Panditi Sarayu	Full related work survey, analysis and discussion sections, introduction and conclusion writing, report editing and proofreading.

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